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The Relationship Between Economic Growth & Income Equality in South Asia

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Abstract

This study analyzes whether economic growth eventually reduces income inequality in South Asian countries, testing the long-run implications of the Kuznets hypothesis. Using panel data for five countries: Pakistan, Nepal, India, Bangladesh, and Sri Lanka. Inequality is measured through the Palma ratio, a metric that better captures disparities between the richest and poorest groups.

The model includes key economic and institutional variables which is expected to play a major role. The findings aim to determine whether growth alone can reduce inequality or if deeper structural issues make inequality more persistent and difficult to overcome.

Methodology

Data: This study uses panel data from five South Asian countries covering 1994–2024.

Missing Data: Missing values were handled using multiple imputation. Twenty imputed datasets were created for each country, along with twenty pooled all-country datasets. Datasets were combined using Rubin's Rule to account for uncertainty across imputations.

Regression Model: The main analysis uses an OLS regression with country and year fixed effects. Country fixed effects control for differences across countries, while year fixed effects control for shared regional or global shocks.

$$Palma_{it} = \alpha + \beta_1 GDPGrowth_{it} + \gamma X_{it} + \mu_i + \lambda_t + \varepsilon_{it}$$

Controls: The model includes education, foreign direct investment, GDP per capita, industry share, inflation, trade, unemployment, urbanization, voice and accountability, control of corruption, and government effectiveness.

Hypothesis

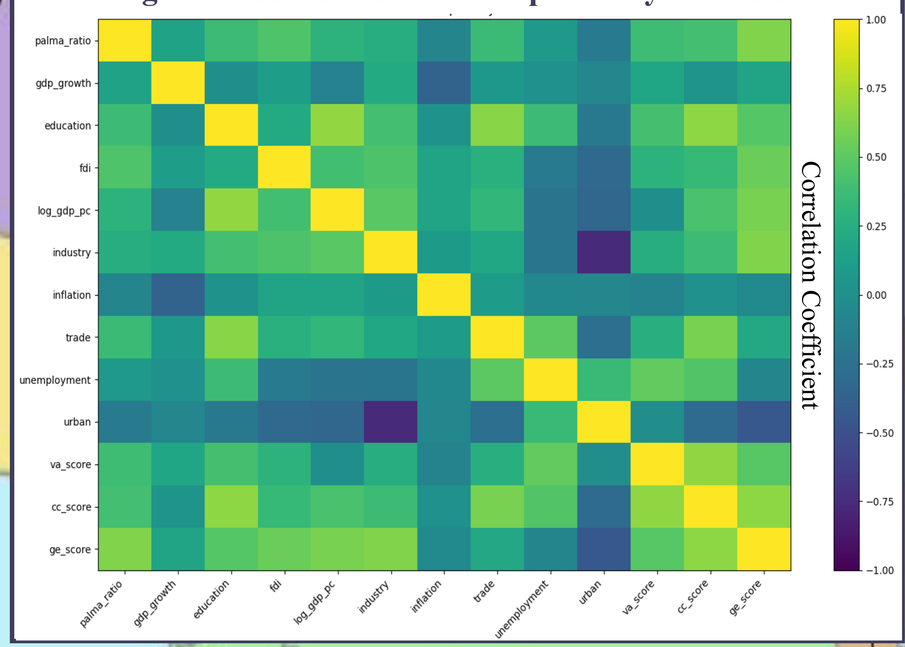
H_0: Economic growth has no effect on income inequality in the long run.

$$H_0: \beta_{growth} = 0$$

H_α: Economic growth reduces income inequality in the long run.

$$H_\alpha: \beta_{growth} < 0$$

Figure 1. Correlation Heatmap for Key Variables



Findings

I. GDP growth has a negative relationship with income inequality across imputed datasets

II. Negative GDP growth coefficient of -0.0300 indicates higher growth is liked to lower inequality, supporting the Kuznets hypothesis

III. The p-value is 0.0496. Therefore, we reject the null hypothesis at the 5% significance level.

IV. The 95% confidence interval is [-0.0600, -0.0001], which is barely below zero.

V. Because the p-value is very close to 0.05, the evidence is statistically significant but modest, so the result should be interpreted carefully.

Figure 2. Relationship Between GDP Growth and Palma Ratio

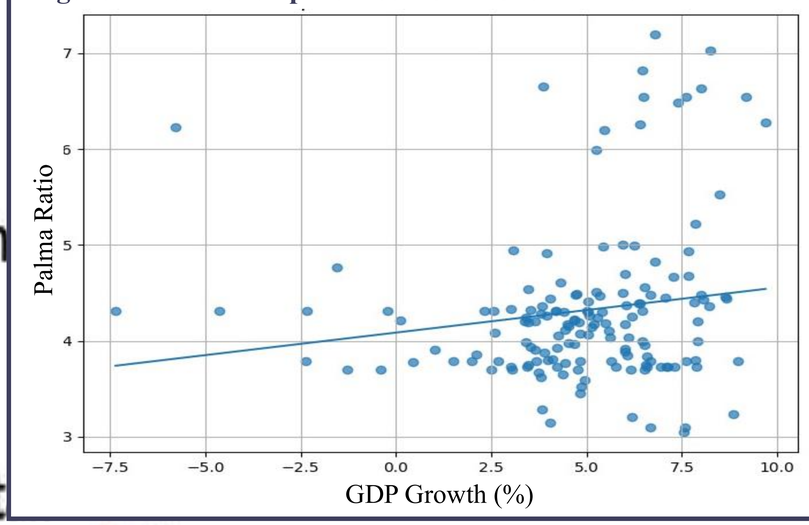
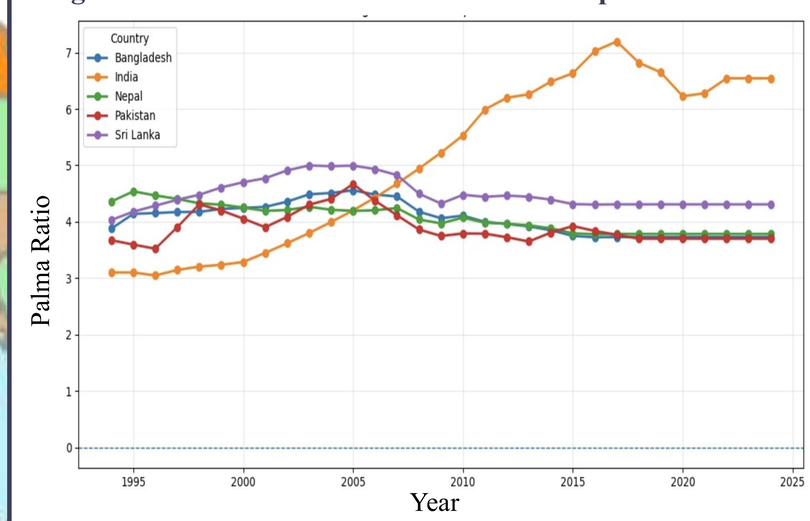


Figure 3. Palma Ratio Trends Across 20 Imputed Datasets



What's Next?

I. Test short-run (5–10 years) vs long-run (20–30 years) effects of growth on inequality

II. Apply additional panel models (fixed and random effects) and compare results.

III. Further explore the roles of education, governance, and trade

IV. Expand the dataset to address current data limitations.

V. Conduct deeper country-by-country analysis to capture heterogeneity